# Department of Statistics

Savitribai Phule Pune University

ST-402 Project

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***To Evaluate the Factors Influencing the Prevalence of Anemia in Women and Children***

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# 1. Introduction

Anemia is a significant public health issue, especially among women and children in India. It is caused by low levels of haemoglobin in the blood, leading to fatigue, weakness, and impaired physical and cognitive development.

The prevalence of anemia is influenced by multiple factors such as poor nutrition, low socio-economic status, lack of education, and limited access to healthcare and sanitation. Despite various health initiatives, anemia continues to affect a large portion of the population.

This project aims to analyse the factors influencing anemia in women and children using data from the National Family Health Survey (NFHS). By applying statistical analysis and machine learning techniques, the study identifies key determinants of anemia and provides insights that can support effective public health interventions.

# 2. Objective

The main objectives of this project are:

- To analyse the prevalence and severity of anemia in women and children
- To examine the relationship between anemia and socio-economic and demographic factors such as education, wealth index, and place of residence
- To identify key factors influencing anemia using statistical analysis
- To apply machine learning models to predict anemia levels
- To derive meaningful insights that can support public health interventions.

# 3. Data Description

The dataset used in this study is obtained from the National Family Health Survey (NFHS), a large-scale survey conducted across India. It contains comprehensive information on demographic, socio-economic, and health-related characteristics of women and children.

The dataset consists of a large number of observations with multiple variables related to age, education, wealth index, place of residence, nutritional factors, and healthcare access. The target variables considered in this analysis are anemia levels in women and children, categorized into severe, moderate, mild, and non-anaemic groups.

Before analysis, the dataset was pre-processed by handling missing values and selecting relevant variables. A cleaned subset of the data was used for further statistical analysis and machine learning modelling. The dataset provides a reliable basis for analysing patterns and identifying key factors influencing anemia prevalence. The large sample size of the dataset enhances the reliability and robustness of the analysis.

# 4. Methodology

## 4.1 Data pre-processing

The dataset was pre-processed to ensure accuracy and reliability of analysis. Missing values were handled appropriately, and inconsistent entries were removed. Relevant variables such as age, education level, wealth index, and anemia status were selected for analysis. The dataset was further cleaned and formatted to make it suitable for statistical analysis and machine learning models.

## 4.2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the distribution of anemia levels and to identify relationships between anemia and various demographic and socio-economic factors.

1. Anemia Distribution

The distribution of anemia levels among the population is shown below:

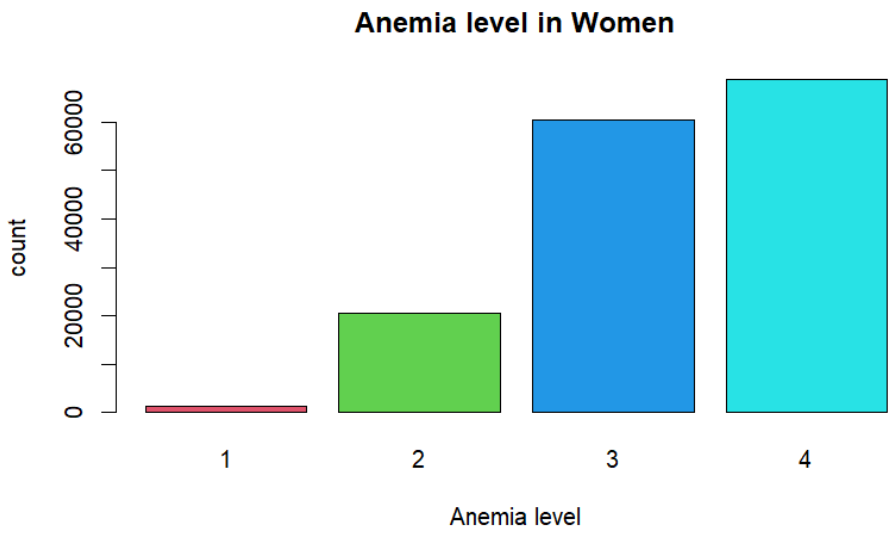


*Figure 1: Distribution of Anemia Levels*

The figure shows that a significant portion of the population falls under mild and moderate anemia categories, indicating a high prevalence of anemia.

2. Place of Residence

The distribution of the population based on place of residence (urban and rural) is shown below:

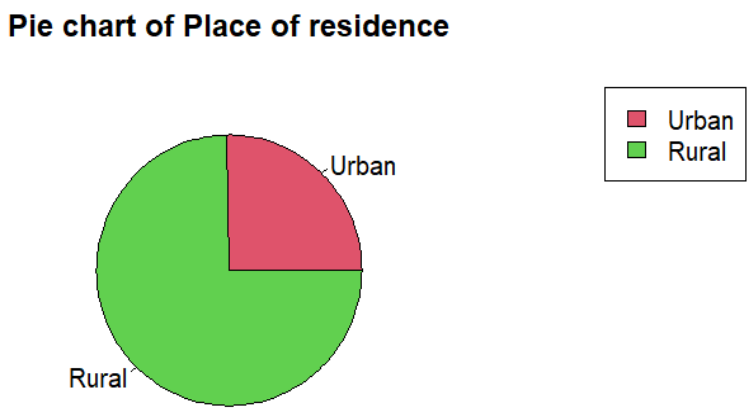


*Figure 2: Distribution by Place of Residence*

The graph indicates that a larger proportion of the population resides in rural areas, which may be associated with higher anemia prevalence due to limited access to healthcare and nutrition.

3. Education Level Distribution

The distribution of educational attainment is presented below:

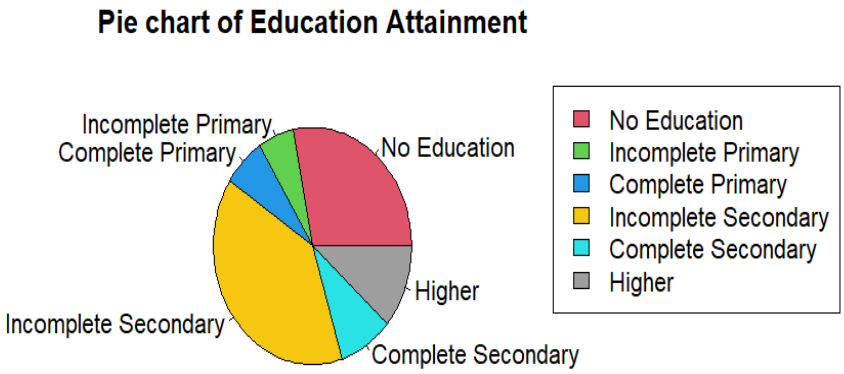


*Figure 3: Distribution of Educational Attainment*

The figure shows that a considerable portion of the population has lower levels of education. Lower education levels are often associated with reduced awareness of nutrition and health practices.

## 4. Wealth Index vs Anemia Level

The distribution of wealth index vs Anemia Level is shown below:

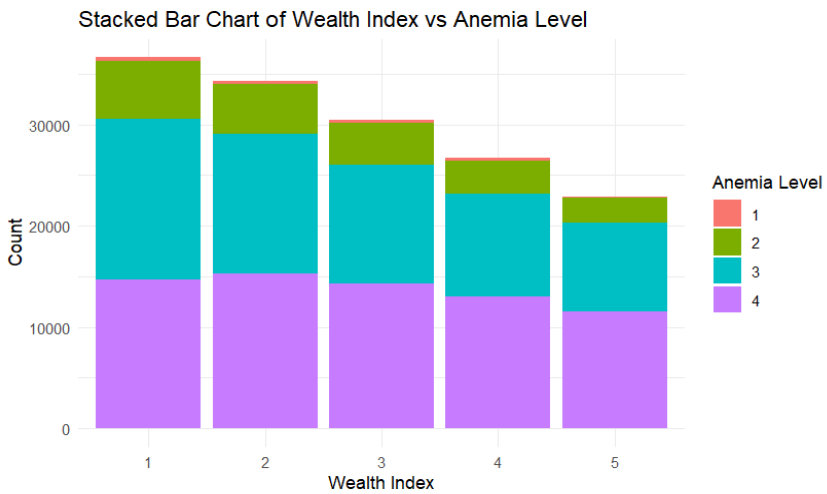


*Figure 4: Distribution of Wealth Index vs Anemia Levels*

The stacked bar chart illustrates the relationship between wealth index and anemia levels. It can be observed that individuals belonging to lower wealth categories have a higher proportion of anemia cases compared to those in higher wealth groups.

The results indicate that individuals from lower wealth categories are more vulnerable to anemia, highlighting the strong influence of economic status on health outcomes.

5. Educational Attainment vs Anemia Level

The distribution of education level vs Anemia is shown below:

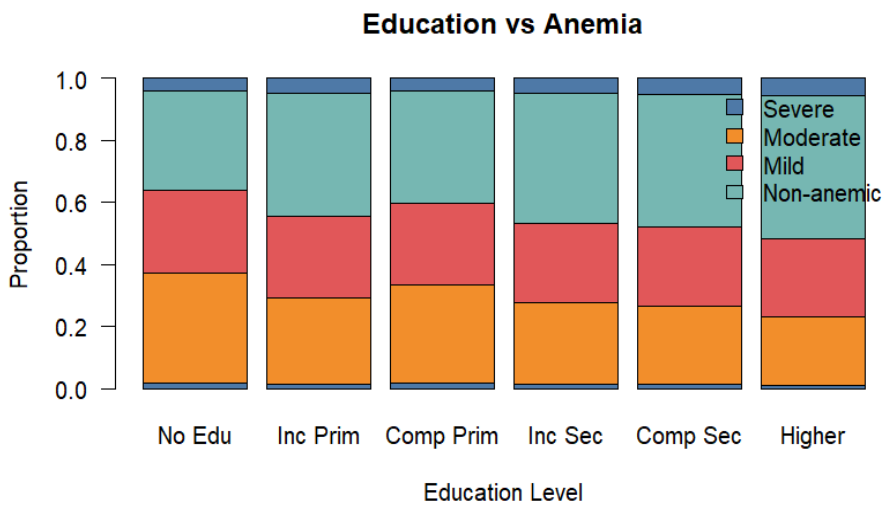


*Figure 5: Distribution of Educational Attainment vs Anemia Levels*

The stacked bar chart illustrates the distribution of anemia levels across different categories of educational attainment. Each bar represents the proportion of individuals within a specific education level classified into anemia categories (severe, moderate, mild, and non-anaemic).

It can be observed that individuals with lower levels of education exhibit higher proportions of moderate and severe anemia, while higher education levels are associated with increased proportions of non-anaemic individuals, indicating improved health outcomes.

Overall, the chart indicates a negative relationship between education level and anemia severity, where higher education is linked to reduced prevalence and severity of anemia.

6. Place of Residence vs Anemia Levels

The distribution of education level vs Anemia is shown below:

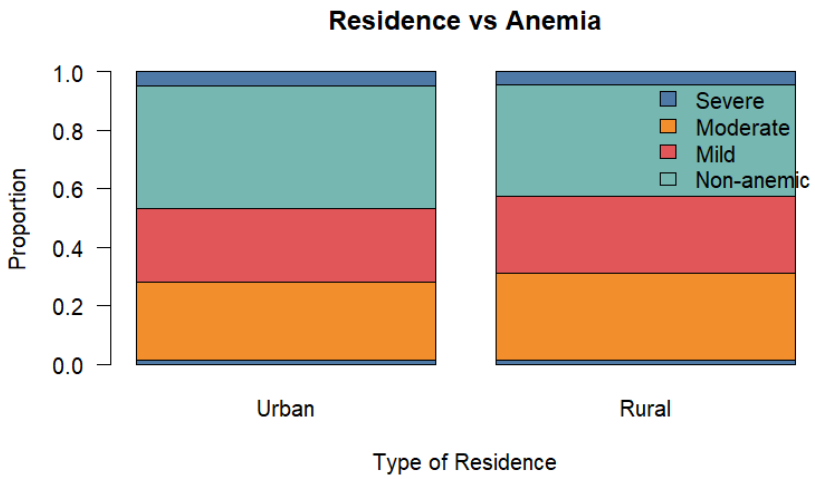


*Figure 6: Distribution of Place of Residence vs Anemia Levels*

The stacked bar chart shows the distribution of anemia levels across different types of residence (urban and rural). Each bar represents the proportion of individuals within each residential category classified into anemia levels (severe, moderate, mild, and non-anaemic).

The proportion of severe anemia is low in both urban and rural areas, with minimal variation between the two groups. Overall, while the differences are not very large, the chart suggests that individuals in urban areas tend to have slightly better health outcomes compared to those in rural areas. This may be due to better access to healthcare and nutrition in urban areas.

## 4.3 Statistical Analysis

Statistical analysis was conducted to examine the relationship between anemia levels and various categorical variables. The Chi-Square test was applied to determine whether there is a significant association between anemia and factors such as education level, wealth index, place of residence, and sanitation.

The results indicated that these variables have a statistically significant association with anemia levels (p-value < 0.05), suggesting that socio-

economic and demographic factors play an important role in determining anemia prevalence.

## 4.4 Machine Learning

Machine learning models were applied to predict anemia levels and identify important contributing factors.

Logistic Regression was used as a baseline classification model to estimate the probability of anemia. Decision Tree was applied to understand decision rules and identify key influencing features. Random Forest, an ensemble learning method, was used to improve predictive performance and handle complex relationships in the data.

These models were implemented to compare their effectiveness in predicting anemia levels and to determine the most suitable approach.

## 4.5 Model Evaluation

The performance of the machine learning models was evaluated based on their predictive accuracy and ability to correctly classify anemia levels.

Among the models, Random Forest demonstrated the highest predictive performance due to its ability to reduce overfitting and capture non-linear relationships. Logistic Regression provided a baseline for comparison but was limited in handling complex patterns. The Decision Tree model offered interpretability but showed signs of overfitting.

Overall, the evaluation indicates that Random Forest is the most suitable model for predicting anemia levels in this study.

# 5. Results and Findings

The analysis revealed that socio-economic status plays a significant role in determining anemia levels among individuals. It was observed that individuals belonging to lower wealth categories are more prone to anemia compared to those in higher wealth groups.

Nutritional intake was also identified as a key factor influencing anemia prevalence. Poor dietary habits and lack of essential nutrients contribute significantly to higher anemia levels, especially among vulnerable populations.

Furthermore, machine learning models such as Logistic Regression, Decision Tree, and Random Forest were used to identify important predictors of anemia. Among these, Random Forest provided better performance and helped in understanding the relative importance of different factors affecting anemia.

Overall, the results indicate that anemia is influenced by a combination of socio-economic, nutritional, and demographic factors, highlighting the need for targeted interventions. These findings are consistent with the patterns observed during exploratory data analysis and statistical testing.

# 6. Conclusion

The study highlights the significant impact of socio-economic and nutritional factors on anemia prevalence among women and children. It was observed that individuals from lower wealth groups and with limited access to proper nutrition are more likely to suffer from anemia.

The analysis also demonstrated that factors such as education, place of residence, and living conditions play an important role in influencing anemia levels. Machine learning models further supported these findings by identifying key predictors of anemia.

The study emphasizes the need for targeted interventions focusing on improving nutrition, education, and healthcare accessibility to effectively reduce anemia prevalence.